

Reading Line Plots Marked in Fractions

Answer Key

Grade 4–5

Quick Answers

1. 1	5. 3	9. 3
2. $\frac{3}{8}$	6. $\frac{3}{8}$	10. $\frac{15}{2}$
3. 3	7. 4	11. $\frac{3}{4}$
4. 8	8. 1	12. 2

Curriculum

Level: Grade 4–5

4.MD / 5.MD – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

5. *If they got 8: counted the X's instead of adding the amounts they stand for*

10. *If they got 10: counted the leaves instead of adding the lengths they stand for*

What this sheet is teaching, and what to listen for. Everything here turns on one distinction: an X is a *thing*, and the fraction under it is a *size*. “How many” questions are answered by counting X’s and always come out as whole numbers. “How much” questions are answered by adding the fractions the X’s sit on and almost never do. When an answer looks wrong, the fastest diagnostic is to ask the child whether their answer counts beakers or measures water — most errors on this sheet are the two swapped.

1. Count one stack. How many beakers held exactly $\frac{1}{2}$ cup?

The fourth mark along the bottom is $\frac{1}{2}$. Counting straight up from it there is a single X. 1

The answer is a beaker, not an amount. A child who writes $\frac{1}{2}$ has repeated the label instead of counting the stack. Worth catching here, on the easiest problem, because every later question depends on the difference.

2. The tallest stack. Which amount was measured most often?

Compare the heights: $\frac{1}{8}$ has one X, $\frac{1}{4}$ has one, $\frac{3}{8}$ has four, $\frac{1}{2}$ has one, $\frac{5}{8}$ has one. The tallest stack sits above $\frac{3}{8}$. $\frac{3}{8}$ cup

Four beakers held that amount — half of all the beakers on the plot.

Two questions, two kinds of answer. “Which amount” wants a fraction from the bottom of the plot ($\frac{3}{8}$); “how many beakers” wants the height of that stack (4). Both point at the same tall column. Ask which one the child wrote.

3. How many more? How many more beakers held $\frac{3}{8}$ cup than $\frac{1}{4}$ cup?

Above $\frac{3}{8}$ there are four X's; above $\frac{1}{4}$ there is one.

$4 - 1 = 3$, so three more beakers held $\frac{3}{8}$ cup.

3

If they answered $\frac{1}{8}$: they subtracted the two fractions printed under the line ($\frac{3}{8} - \frac{1}{4} = \frac{1}{8}$) instead of the two stack heights. That subtraction answers a different, perfectly good question — “how much more water was in one beaker than the other?” — and it is worth asking the child which of the two questions was on the page.

4. How many beakers?

Count the X's stack by stack: $1 + 1 + 4 + 1 + 1 = 8$ beakers.

8

A check worth teaching. There are five marks along the bottom and eight X's above them. A child who answers 5 has counted the marks — the amounts that were *possible* — rather than the beakers that were *measured*.

5. How much water? How much water is there altogether, in cups?

Now the fractions themselves have to be added, one for every X. Rewrite them all in eighths first: $\frac{1}{4} = \frac{2}{8}$ and $\frac{1}{2} = \frac{4}{8}$.

$$\frac{1}{8} + \frac{2}{8} + \frac{3}{8} + \frac{3}{8} + \frac{3}{8} + \frac{3}{8} + \frac{4}{8} + \frac{5}{8} = \frac{24}{8}$$

And $\frac{24}{8}$ is 24 eighths, which is exactly 3 whole cups.

3 cups

If they answered 8: they counted the X's instead of adding the amounts. Eight beakers, three cups — both are true, and they answer different questions. Pouring is the image that helps: eight beakers going into one jug does not make eight cups of water.

6. Share it out. How much water would each beaker hold?

The jug holds 3 cups and there are 8 beakers, so the question is $3 \div 8$.

Three cups shared eight ways: each whole cup splits into eight eighths, giving 24 eighths in total, and $24 \div 8 = 3$ eighths each.

 $\frac{3}{8}$ cup

This is the idea behind fair shares. Notice that $\frac{3}{8}$ is also the amount that came up most often — a coincidence on this plot, not a rule. The equal-share amount is what every beaker *would* hold if the water were levelled out; it need not be an amount any beaker actually held.

If they answered $\frac{3}{5}$: they divided by the five marks on the number line instead of by the eight beakers. Marks are amounts, beakers are things — the same swap as problem 4, one step further on.

7. A new plot. How many leaves were $\frac{3}{4}$ inch long?

On the leaf plot the marks count by quarters. Above the $\frac{3}{4}$ mark there are four X's.

4

Two leaves were exactly one inch long.

Read the scale before the stacks. The first plot counted by eighths and this one counts by quarters, so the same physical distance along the line now means twice as much length. Children who do not check the scale read the second plot with the first plot's ruler.

8. Longest and shortest. How much longer was the longest leaf than the shortest one?

The furthest-right X sits above $1\frac{1}{4}$ inches; the furthest-left X sits above $\frac{1}{4}$ inch.

$1\frac{1}{4} - \frac{1}{4} = 1$, so the longest leaf was one inch longer than the shortest.

1 in

This answer is a length, so it takes a unit. Unlike problems 1, 3, 4 and 7, both numbers being subtracted came from the *bottom* of the plot rather than from the stacks, which is exactly why the answer is measured in inches instead of counted in leaves.

9. Longer than three quarters. How many leaves were longer than three quarters of an inch?

Everything to the right of the $\frac{3}{4}$ mark counts: the mark at one inch carries two X's and the mark at $1\frac{1}{4}$ inches carries one.

$2 + 1 = 3$ leaves.

3

“Longer than” does not include the mark itself. The four X's sitting *on* $\frac{3}{4}$ are leaves that are exactly three quarters of an inch long, so they are not longer than it. A child who answers 7 has swept those four in — ask them whether a leaf can be longer than itself.

10. End to end. How long is the row of leaves?

Add one length per X. In quarters the ten leaves are 1, 2, 2, 3, 3, 3, 3, 4, 4, 5 quarters, and those add to 30 quarters.

30 quarters is $30 \div 4 = 7$ whole inches with 2 quarters left over, and 2 quarters is a half.

$7\frac{1}{2}$ in

If they answered 10: they counted the leaves rather than adding the lengths. Ten leaves, seven and a half inches — the row of leaves on the desk is the picture that settles it.

11. If every leaf were the same. How long would each leaf be?

The row is $7\frac{1}{2}$ inches long and there are ten leaves, so the question is $7\frac{1}{2} \div 10$.

In quarters: 30 quarters shared among ten leaves is 3 quarters each.

$\frac{3}{4}$ in

Worth pausing on. Three quarters of an inch is also the commonest leaf length on the plot, so a child can arrive at the right answer for the wrong reason. Ask them to say which they did: shared the total out, or read the tallest stack. Only the first one is what the question asked, and on a different plot the two answers come apart.

If they answered $1\frac{1}{2}$: they divided by the five different lengths on the line instead of by the ten leaves.

12. Two different questions.

(a) Above the $\frac{1}{2}$ mark there are two X's, so the number of leaves that were half an inch long is 2

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(b) (Open explanation — listen to the reason.)

A model answer. “To answer *how long is one leaf*, you look along the bottom of the plot, because the fractions there are the lengths. To answer *how long are all the leaves together*, you have to add up one fraction for every X, because each X is a leaf and it brings its own length with it. Counting the X's only tells you how many leaves there are, and ten leaves could be ten tiny ones or ten long ones, so counting can never tell you how long the row is.”

Marking guide. Full credit for any answer that says lengths are read from the bottom of the plot while the total needs the fractions added, and that gives a reason why a count alone is not enough. Partial credit for correctly naming which part answers each question with no reason attached. Not sufficient: “because you add for one and count for the other” — a restatement of the method rather than of why it works.

If the child is stuck: ask them to imagine a second plot with ten X's all sitting on $1\frac{1}{4}$. It has the same number of X's as Ravi's plot and a much longer row, which makes the point without any arithmetic.

Open explanation — see model answer.